



Direct determination of permeability, diffusivity and solubility of gases in membrane based on simultaneous monitoring of pressure rise and decay in a single time-lag experiment

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ARTICLE INFO

Keywords:

Membrane characterization
Constant volume systems
Time-lag method
Pressure decay
Differential pressure transducer

ABSTRACT

The time-lag method is the most common approach for evaluating the membrane gas permeability, diffusion and solubility coefficients in a single gas permeation experiment. However, only permeability and diffusivity are determined directly, while the solubility is evaluated from the ratio of permeability and diffusivity. Herein, we present how the conventional time-lag experiment can be used to determine all three transport coefficients directly when, in addition to monitoring the pressure rise in the constant volume downstream from the membrane, the corresponding pressure decay is also monitored in the constant volume upstream of the membrane. The new design of the upstream part of the constant volume (CV) system used in this study allowed pressure decay monitoring. We provide the mathematical fundamentals of the new methods enabled by the new CV system and demonstrate them experimentally using a commercial poly (dimethylsiloxane) PDMS membrane.

1. Introduction

Membrane gas separation became an industrial reality in the late 1970s with Monsanto's first commercial asymmetric hollow membrane modules [1]. Today's prominent membrane-based gas separations include hydrogen recovery [2], natural gas purification [3], and air separation [4]. In addition, olefin/paraffin separation, ethanol/water separation, and carbon capture applications have expanded the opportunities for gas separation membranes [5]. Despite significant progress in the development of inorganic [6] and mixed matrix membranes (MMMs) [7], commercial gas separation membranes are exclusively prepared using organic polymers. These include polysulfones, cellulose acetates, polyphenylene oxide, aramids, polycarbonates, and polyimides [5]. Developing new materials - organic, inorganic, and composite - remains the central theme of membrane gas separation research. The combination of permeability and permselectivity for a given pair of gases compared to the upper-bound line is the standard metric for assessing new membrane materials [8]. In turn, the permeability (P) and permselectivity for a given pair of gases depend on the respective diffusivity (D) and solubility (S) coefficients of these gases in the membrane material.

The three transport coefficients, P , D , and S , are determined in a

single dynamic gas permeation experiment, often referred to as a time-lag experiment. The mathematical basis for the time-lag experiment was first described more than a century ago [9]. The time-lag experiments are generally performed in constant volume (CV) systems. A degassed, homogeneous membrane from the tested material is exposed to a step pressure increase at one (feed) side. The resulting pressure rise at the other (permeate) side of the membrane is monitored in real-time. The tangent of the linear portion of the pressure rise, associated with a constant permeation rate, is directly proportional to P [10]. The extrapolation of that linear portion to the time axis yields a time lag (θ_d), which is related to the diffusivity by:

$$\theta_d = \frac{L^2}{6D} \quad (1)$$

where L is the membrane thickness. According to the solution-diffusion model, the ratio of P to D represents S . The linear portion of the pressure response indicates a pseudo-steady state permeation, which is commonly referred to as a steady state.

Alternatively, gravimetric, barometric, and volumetric techniques can determine S directly in gas sorption experiments [11]. The outcome of gas sorption experiments is an adsorption isotherm, i.e., the relation between the gas concentration in the membrane (c) as a function of

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<https://doi.org/10.1016/j.memsci.2025.123741>

Received 29 November 2024; Received in revised form 2 January 2025; Accepted 18 January 2025

Available online 23 January 2025

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pressure (p) and solubility is

$$S = \frac{c}{p} \quad (2)$$

The solubility might be independent of pressure, like in the case of rubbery polymer membranes, in which Henry's law governs gas sorption. However, typically, S depends on p . For example, in the case of glassy polymer membranes, which represent most commercial gas separation membranes [5], gas sorption is governed by dual-mode sorption [12]. Also, gas sorption in novel membrane materials, polymeric, inorganic, and mixed matrices does not follow Henry's law and is not necessarily dual-mode sorption. Gas sorption experiments rely on reaching an equilibrium between the gas and the membrane phases, which may require a long time, especially for thick membranes, i.e., when the characteristic length for gas diffusion in a sample is relatively large. On the other hand, monitoring the gas concentration in the membrane as a function of time also allows the direct determination of the gas diffusivity. Knowing D and S , the gas permeability is evaluated from the product of D and S [13,14]. Therefore, like the time-lag method, the dynamic sorption/desorption experiments allow determining all three transport coefficients from a single experiment. However, two (S and D) are determined directly, while the third (P) is determined indirectly.

The dynamic sorption/desorption experiments are usually performed using a barometric method called the pressure decay method. This technique introduces a known amount of gas into an evacuated chamber containing the membrane sample, and the pressure decay in the chamber corresponds to gas sorption by the membrane. Therefore, this method relies on the accurate knowledge of the system volume, a sensitive pressure transducer, and maintaining a constant temperature [11].

The protocol of a pressure decay method resembles that of the time-lag method from the perspective of the upstream of the tested membrane. The main difference is that the steady-state permeation is reached in the time-lag method rather than an equilibrium between the gas and the membrane. However, it should be emphasized that the time-lag method relies on the instantaneous equilibrium between the gas and the membrane's upstream interface. Another difference is that the pressure decay method is commonly used in gas sorption/desorption experiments, but its equivalent in gas permeation does not practically exist, even though the benefits of applying the pressure decay method in membrane characterization were first discussed 30 years ago [15]. Monitoring the upstream pressure decay can offer some unique advantages, particularly when characterizing barrier materials for which reaching the steady state permeation (necessary to determine the time lag) could take not seconds or minutes but hours or even days. Combining the upstream pressure decay with the downstream pressure rise allowed Al-Ismaïly et al. to determine P from transient permeation data before reaching a steady state [16]. Immediately after initiating the gas permeation experiment, the membrane behaves as a semi-infinite solid. It can be shown that during that time, the pressure decay (Δp_0) is a linear function of the square root of time [16]:

$$\Delta p_0 = -\frac{2p_0 S R T A \sqrt{D}}{V_0 \sqrt{\pi}} \sqrt{t} \quad (3)$$

where p_0 is the upstream (feed) pressure, V_0 is the constant upstream volume, A is the membrane area, T is the absolute temperature, and R is the universal gas constant. Eq. (3) implies that $\Delta p_0 \ll p_0$. The slope of Δp_0 vs $\sqrt{t}$ is directly proportional to $S\sqrt{D}$, but the intercept from Eq. (3) is zero. Therefore, S and D cannot be determined from the initial pressure decay. However, it was shown that laminating the tested membrane with a film of known properties (e.g. silicone rubber) results in an intercept that depends on the tested membrane's S and D [17]. Therefore, the mathematical expression for the intercept yields the second independent equation, thus allowing the determination of D , S , and P

from the upstream pressure decay before the gas leaves the tested barrier material. It can also be shown that if the tested membrane is not laminated and the gas emerges from the membrane, the deviation from the initial linear relation given by Eq. (3) at any time depends on D but not S [18]. Therefore, D can be evaluated from the observed deviation at the specific time, and knowing D , S can be determined from the initial slope of Δp_0 versus $\sqrt{t}$.

Despite the potential advantages of monitoring the pressure decay in membrane characterization, this approach did not gain much attention in the membrane community. There are two main reasons for that. First, the resolution of the absolute pressure transducers depends on the maximum pressure they can measure. Because the pressure at the upstream side of the membrane is orders of magnitude greater than that at the permeate side, the pressure rise in the conventional time-lag method can be monitored with an excellent resolution, but not the pressure decay at the upstream side of the membrane. Another challenge when monitoring pressure decay is the temperature drift [19]. The measurements in a CV system assume the observed pressure rise or decay (dp/dt) depends only on the gas molar flow (dn/dt) into or out of the membrane. It is a reasonable assumption when monitoring the pressure rise starting at high vacuum ($p \approx 0$), but not necessarily when monitoring the pressure decay after initiating a time-lag experiment, which is evident when considering the total derivative from the ideal gas law:

$$\frac{dp}{dt} = \frac{RT}{V} \frac{dn}{dt} - \frac{p}{V} \frac{dV}{dt} + \frac{p}{T} \frac{dT}{dt} \quad (4)$$

The second term on the right-hand side is always zero. However, when monitoring the pressure decay, the last term on the right-hand side might not be negligible even when the temperature is controlled because of the magnitude of the feed pressure ($p \gg 0$). The resolution problem of absolute pressure transducers can be solved by adopting the concept of a two-tank volume, initially developed by Arkilic et al. [20], commonly used in microfluidics. Al-Ismaïly et al. [16] were the first to adopt this concept to measure pressure decay in membrane characterization. However, the time-lag experiments could not be performed at pressures greater than 175 Torr. Moreover, the upstream and downstream time lag ratio did not yield a reasonable value, attributed to a "compression effect." [16]. We have recently solved the problems faced by Al-Ismaïly et al. by developing a new generation of the CV system [21], which increased the range of the feed pressure to 2000 Torr and led to the absolute ratio of the time lags for PDMS membrane very close to the expected value of 2.

Like the conventional time-lag technique, the new characterization methods discussed in Refs. [16–18], which rely on the accurate monitoring of the pressure decay, allow the direct determination of two out of three primary gas transport coefficients in the membrane, namely D and P . In this study, we present a new method based on the conventional time lag, which relies on the simultaneous monitoring of pressure rise and decay and allows the direct determination of all three transport coefficients from a single gas permeation experiment. We first provide the mathematical fundamentals of the new method. Then, we experimentally demonstrate the application of this method in the new CV system using a commercial PDMS membrane. We also discuss how the method proposed in this study and the new generation of CV systems can facilitate the characterization of new membrane materials for gas separation.

2. Theoretical background

The governing partial differential equation (PDE) describing the time-lag experiment is the Fick's second law of diffusion:

$$\frac{\partial c(x,t)}{\partial t} = D \frac{\partial^2 c(x,t)}{\partial x^2} \quad (5)$$

The corresponding initial and boundary conditions are: $c(x, 0) = 0$; $c(0, t) = c_0$; $c(L, t) = 0$. Solving the governing PDE, subject to the above

initial and boundary conditions, using the separation of variable method leads to:

$$c(x, t) = c_0 \left(1 - \frac{x}{L}\right) - 2 \frac{c_0}{\pi} \times \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right) \exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) \quad (6)$$

Recognizing that $c_0 = p_0 S = p_0(P/D)$, where p_0 is the upstream pressure after the step change to initiate the experiment, Eq. (6) can be re-written as:

$$c(x, t) = \frac{p_0 P}{D} \left(1 - \frac{x}{L}\right) - 2 \frac{p_0 P}{\pi D} \times \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right) \exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) \quad (7)$$

Using Fick's first law allows the evaluation of the gas flux anywhere within the membrane:

$$J(x, t) = -D \frac{dc(x, t)}{dx} = \frac{p_0 P}{L} + 2 \frac{p_0 P}{L} \times \sum_{n=1}^{\infty} \cos\left(\frac{n\pi x}{L}\right) \exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) \quad (8)$$

Evaluating Eq. (8) at $x = 0$ and $x = L$ allows the determination of the time-dependent gas flux in and out of the membrane:

$$J(0, t) = \frac{p_0 P}{L} + 2 \frac{p_0 P}{L} \times \sum_{n=1}^{\infty} \exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) \quad (9)$$

$$J(L, t) = \frac{p_0 P}{L} + 2 \frac{p_0 P}{L} \times \sum_{n=1}^{\infty} (-1)^n \exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) \quad (10)$$

The above-series solutions simplify to single-term approximation when the Fourier number (dimensionless time) is relatively large. More specifically, when $Fo = \frac{Dt}{L^2} > 0.2$, Eqs. (9) and (10), become:

$$J(0, t) = \frac{p_0 P}{L} + 2 \frac{p_0 P}{L} \exp\left(-\frac{D\pi^2 t}{L^2}\right) \quad (11)$$

$$J(L, t) = \frac{p_0 P}{L} - 2 \frac{p_0 P}{L} \exp\left(-\frac{D\pi^2 t}{L^2}\right) \quad (12)$$

The difference between the fluxes in and out are:

$$\Delta J(t) = J(0, t) - J(L, t) = 4 \frac{p_0 P}{L} \exp\left(-\frac{D\pi^2 t}{L^2}\right) \quad (13)$$

As $t \rightarrow \infty$, i.e., at steady state, ΔJ approaches to 0. However, in the transient state, when $Fo > 0.2$, taking the natural logarithm of both sides of Eq. (13) leads, after rearrangements, to:

$$\ln\left\{\frac{\Delta J(t)L}{4p_0 P}\right\} = -\frac{D\pi^2}{L^2} t \quad (14)$$

$\Delta J(t)$ can be evaluated by the differentiation of experimentally measured pressure rise and decay; P can be determined from the steady-state pressure rise or decay, while p_0 and L are known parameters. Therefore, the left-hand-side of Eq. (14) is known and can be plotted as a function of time, which should yield a straight line with the slope (σ):

$$\sigma = -\frac{D\pi^2}{L^2} \quad (15)$$

Knowing σ , D can be evaluated from Eq. (15), and it should be the same as D from Eq. (1). Also, monitoring the pressure decay also allows the measurement of the upstream time lag (θ_0), which is given by:

$$\theta_0 = -\frac{L^2}{3D} \quad (16)$$

The experimentally measured pressure decay also allows another way to determine D . At early times, i.e., before permeating molecules emerge from the permeate interface of the membrane, the membrane behaves as semi-infinite, and the pressure decay is given by Eq. (3). In this early transient permeation period, the plot of the pressure decay versus the square root of time should yield a straight line with the slope (Σ_t) given by:

$$\Sigma_t = -\frac{2p_0 SRTA\sqrt{D}}{V_0\sqrt{\pi}} \quad (17)$$

On the other hand, when gas permeation reaches a steady state, the upstream pressure becomes a linear function of time, and the corresponding steady-state slope (Σ_{ss}) is given by:

$$\Sigma_{ss} = -\frac{SDp_0ART}{V_0L} \quad (18)$$

The ratio of the two slopes leads to the following expression:

$$\frac{\Sigma_{ss}}{\Sigma_t} = \frac{\sqrt{\pi}}{2L} \sqrt{D} \quad (19)$$

Therefore, the ratio of the two slopes allows for determining D directly from Eq. (19).

Eqs. (1), (15), (16) and (19) provide four ways to evaluate D from the same time-lag experiment, which should lead to the same value. Although more independent ways to determine a given parameter increase confidence in that parameter, the analysis so far does not go beyond the conventional time-lag method because only D and P are determined directly. S needs to be determined from the ratio of P and D . However, the accurate measurement of the pressure rise and decay also allows for the direct determination of S .

First, we recognize that gas accumulation in the membrane as a function of time, $M(t)$, is related to the fluxes in and out of the membrane:

$$M(t) = A \left\{ \int_0^t J(0, t) dt - \int_0^t J(L, t) dt \right\} \quad (20)$$

Substituting Eqs. (9) and (10) into Eq. (20) leads to:

$$M(t) = 2 \frac{Ap_0 SL}{\pi^2} \left\{ \sum_{n=1}^{\infty} ((-1)^n - 1) \frac{1}{n^2} \left(\exp\left(-\frac{Dn^2\pi^2 t}{L^2}\right) - 1 \right) \right\} \quad (21)$$

As $t \rightarrow \infty$, i.e. at steady state, $M(t)$ becomes a constant value M_{∞} :

$$M_{\infty} = 2 \frac{Ap_0 SL}{\pi^2} \sum_{n=1}^{\infty} -\frac{((-1)^n - 1)}{n^2} = 2 \frac{Ap_0 SL}{\pi^2} \left(\frac{\pi^2}{4}\right) = \frac{Ap_0 SL}{2} \quad (22)$$

Knowing the upstream and downstream volumes of the system and the experimentally measured pressure rise and decay, M_{∞} can be evaluated experimentally. In turn, the gas solubility in the membrane becomes:

$$S = \frac{2M_{\infty}}{p_0 LA} \quad (23)$$

It should be emphasized that Eq. (21) also provides the basis for determining gas diffusivity in the membrane. Dividing both sides of Eq. (21) by M_{∞} and rearranging the resulting equation leads to:

$$1 - \frac{M(t)}{M_{\infty}} = \exp\left(-\frac{D\pi^2 t}{L^2}\right) \quad (24)$$

Taking the natural logarithm of both sides of Eq. (24) yields:

$$M_{\ln}(t) = \ln\left(1 - \frac{M(t)}{M_{\infty}}\right) = -\frac{D\pi^2 t}{L^2} \quad (25)$$

The plot of the LHS of Eq. (25) versus time should lead to a straight line with the same slope (σ) as before given by Eq. (15), and then D is determined from Eq. (16).

3. Materials and methods

3.1. Materials

PDMS membranes (SSPM823-010-12"-SSP-M823-Platinum Cured

Silicone) were purchased from Interstate Specialty Products (Sutton, MA). They were supplied with a white virgin Teflon backing. According to the supplier, the PDMS membranes had a $0.025'' \pm 0.002''$ ($0.610 \text{ mm} \pm 0.051 \text{ mm}$) thickness. All experiments were performed using a 99.99 % pure N_2 from the compressed gas cylinder (Linde Canada Inc., Ottawa, ON).

3.2. Apparatus

Fig. 1 presents the schematic diagram of our novel CV system, which has been recently described in great detail [21]. Here, we focus only on the upstream part of the system, which allows the accurate monitoring of the pressure decay. The upstream volume is divided into the working V-101 and reference V-100 sections, with respective volumes of 89 cm^3 and 96 cm^3 . The 96 cm^3 does not include the dead volume (4 cm^3) comprising the upstream part of the membrane cell and tubing up to the remotely operated FCV-7 valve. The working and reference volumes could be connected/separated with another remotely operated valve (FCV-1). The FCV valves comprise a valve body (Swagelok SC-4BK-VCR) and an actuator (Swagelok SS-4BK-VCR-1D & SS-4BK-1C-K10). The bodies of the FV and FCV valves are the same: stainless steel bellows sealed valves, procured from Weston Valve & Fitting LTD, Mississauga, ON. The measurement of the pressure decay due to gas permeation into the membrane relies on a capacitance-based differential pressure transducer DPT1 (CCR Process Products Kanata, Ontario, Canada, MKS698A $\pm 1 \text{ Torr}$, 10^{-6} full-scale resolution, accuracy: 0.05 % of

reading), which compares a variable pressure in V-101 with a constant pressure in V-100. In addition, the working and reference volumes are equipped with identical absolute pressure transducers, APT1 and APT4 (CCR Process Products Kanata, Ontario, Canada, MKS627F13TBC1B, 10^{-5} full-scale resolution, accuracy: 0.12 % of reading, range: 0–2000 Torr). The pressure in the reference volume can be adjusted independently of that in the working volume using a rotary vacuum pump P-400 (CANVACTECH, Ottawa, Ontario, Canada, Edwards RV3, A65201906, ultimate pressure: $1.5 \times 10^{-3} \text{ Torr}$) and compressed gas cylinder TK-400 via FV-6 and FV-3 valves. Except for P-400, a turbomolecular pump P-401 (Edwards Vacuum LLC Sanborn, New York, USA, Edwards TS85D1002, ultimate pressure: $<3.75 \times 10^{-10} \text{ Torr}$) and TK-400, all components of the system in Fig. 1 are enclosed in a temperature-controlled chamber with an operational range between 273 K and 313 K [22].

3.3. Experimental protocol

After installing the membrane inside the cell, the system was extensively evacuated to degas the membrane using two vacuum pumps, P-400 and P-401, and leak tests were performed in the different sections of the system, including the downstream, reference, and working volumes. The observed leak rates (10^{-7} torr/s or lower) were at least four orders of magnitude lower than the measured pressure decay/rise during experiments. After completing the leak tests, the system was evacuated before initiating each experiment. With the membrane cell and the

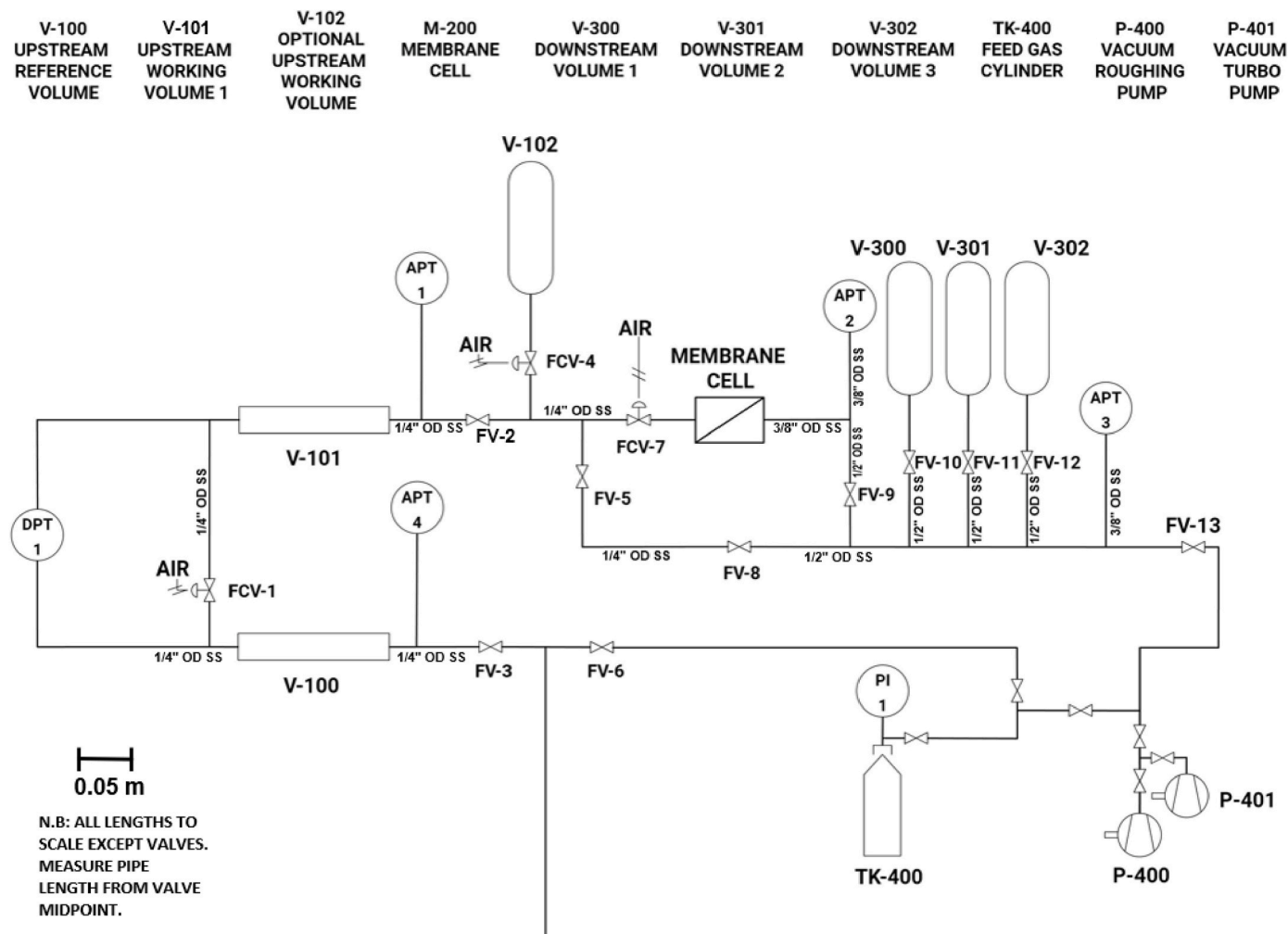


Fig. 1. A schematic diagram (to the scale) of the novel constant volume (CV) system [21].

downstream section of the system under vacuum, the reference and working volumes were pressurized to the desired level using TK-400. The remote opening of FCV-7 initiated the experiment, causing gas expansion and a pressure drop in the working volume greater than 1 torr, which is the full range of DPT1. Consequently, the pressure decay could only be monitored using APT1, not DPT1. For DPT1 to monitor the pressure decay, the pressure in the reference volume was reduced by the expected pressure drop in the working volume before the experiment. However, the adiabatic expansion in the working volume led to a sudden decrease in gas temperature. Since the thermal capacity of the gas in the working volume was negligible compared to the thermal capacity of insulated tubing, the gas temperature quickly returned to its initial level. Due to this phenomenon, the DPT1 readings in the first 5–6 s of the experiment were unreliable and had to be discarded [21].

Consequently, the gas permeation experiments began when the working and reference volumes were connected (i.e., FCV-1 was open), making DPT1 unavailable to monitor the experiment's progress. FCV-1 was closed 5 s after FCV-7 was opened. In other words, the progress of the experiments by DPT1 was monitored after 5 s from their initiation. The remote operation of FCV1 and FCV7 ensured excellent repeatability of the experiments [21].

Simultaneously with the pressure decay, the corresponding downstream pressure rise was monitored using two identical 10-Torr absolute pressure transducers, APT2 and APT3 (CCR Process Products Kanata, Ontario, Canada, MKS627F11TBC1B, 10^{-5} full-scale resolution, accuracy: 0.12 % of reading, range: 0–10 Torr). The gas accumulated in the downstream configuration that excluded accumulation tanks (V-300, V-301, V302), i.e., in the downstream volume of 81 cm³. The purpose of not using the accumulation tanks downstream was to minimize the effect of resistance to gas accumulation, which affects the experimentally measured θ_d [23]. The pressure rise results did not require any processing; the pressure decay results required two corrections described in Ref. [21].

4. Results

Fig. 2 presents an example of the progress of the gas permeation experiment – pressure rise and pressure decay – performed in the CV system shown in Fig. 1. The experiment was performed with a PDMS membrane (0.610 ± 0.051 mm thick) using N₂ at 850 Torr pressure and 30 °C. Because the experiment was started when the working and reference volumes were commented, the pressure decay data is available only after 5 s from the initiation of the experiment, i.e., when FCV-1 was closed. The pressure decay rate is highest at the beginning of the

experiment and gradually decreases as time progresses, reaching a constant rate indicating a steady state. Extrapolating the linear portion of the pressure decay curve to the time axis yields the upstream time lag $\theta_0 = -99$ s. At the same time, right after the initiation of the experiment, there is a negligible pressure rise because it takes time for gas molecules to diffuse across the membrane, which is followed by a gradual increase in the rate of pressure rise until a steady state permeation is attained. The corresponding downstream time lag based on the intercept of the extrapolated linear portion of the pressure rise curve is $\theta_d = 49$ s. Therefore, the corresponding absolute value of the ratio of θ_0 and θ_d is 2.02, which is nearly identical to the theoretical value of 2.

Using θ_d and θ_0 determined from the experimental pressure rise and decay, the corresponding diffusivity of N₂ in PDMS from Eqs. (1) and (16) are 1.26×10^{-9} m²/s and 1.25×10^{-9} m²/s. Based on the slopes of the downstream pressure rise and the upstream pressure decay, the permeability of N₂ in PDMS is 6.83×10^{-14} mol/s·m·Pa (207 Barrer) and 7.30×10^{-14} mol/s·m·Pa (221 Barrer), respectively. Knowing D and P , the corresponding solubility of N₂ in PDMS from the pressure rise and decay is 5.40×10^{-5} mol/m³·Pa and 5.83×10^{-5} mol/m³·Pa, respectively. It is, therefore, evident that D , P , and S from the downstream pressure rise and the upstream pressure decay are very close to each other, which confirms the validity of the experimentally measured pressure decay in our novel CV system.

As shown in Fig. 2, the rise in pressure after the initiation of the experiment is negligible for the first 40 s because it takes time for the permeating molecules to reach the downstream interface of the membrane. During that time, the membrane behaves as a semi-infinite solid, and the pressure decay should be governed by Eq. (3). Fig. 3 presents the experimentally measured pressure decay, shown in Fig. 2, as a function of the square root of time. The inset in Fig. 3 is the pressure decay portion of Fig. 2. As expected, Δp_0 is a linear function of $\sqrt{t}$ at early times. Then, the pressure decay rate gradually increases because the permeating molecules leave the membrane, i.e., the membrane is no longer a semi-infinite solid. The slope of the linear portion Δp_0 versus $\sqrt{t}$ represents Σ_u , given by Eq. (17), and equals -0.065 Torr/s^{0.5}. The corresponding slope of the linear portion of pressure decay in Fig. 2, used to calculate membrane permeability, represents Σ_{ss} , given by Eq. (18), and is equal to 3.4×10^{-3} Torr/s. Therefore, using the ratio of Σ_{ss} and Σ_u , the diffusivity of N₂ in PDMS from Eq. (19) is 1.28×10^{-9} m²/s. This diffusivity, which we will refer to as D by the two-slope method, is very close to the D values from the upstream and downstream time lags, which validates the two-slope method.

The direct determination of the solubility of N₂ in PDMS from Eq.

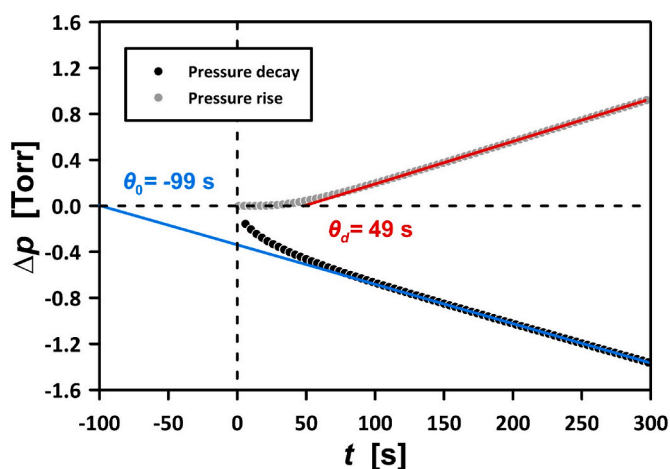


Fig. 2. An example of the pressure profiles from the N₂ permeation in the CV system using a PDMS membrane of thickness 0.610 ± 0.051 mm. The experiment was conducted at 850 Torr and 30 °C.

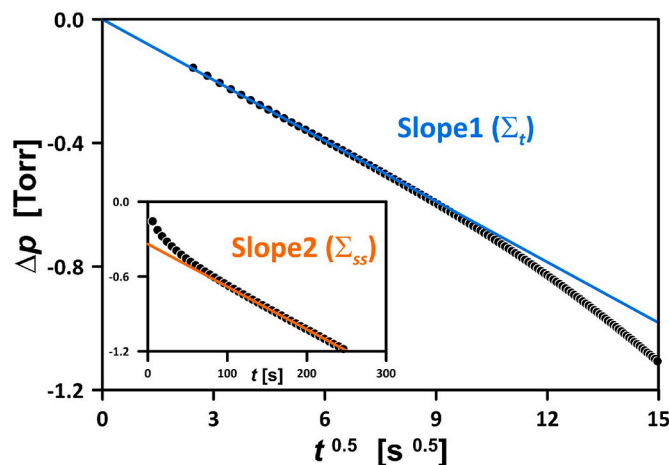


Fig. 3. Pressure decay of the experiment demonstrated in Fig. 2 in the square root of the time domain. The inset shows the pressure decay part of Fig. 2. The N₂ permeation experiment in the CV system using a PDMS membrane of thickness 0.610 ± 0.051 mm at 850 Torr and 30 °C.

(23) requires M_∞ , i.e., the amount of N_2 dissolved in the membrane at a steady state. The number of moles of N_2 in the membrane at a given time, $M(t)$, can be determined from the pressure decay and pressure rise curves shown in Fig. 2. The number of moles entering the membrane, $M_0(t)$, is directly related to the pressure decay:

$$M_0(t) = \frac{V_0}{RT} \int_0^t \Delta p_0(t) dt = A \int_0^t J(0, t) dt \quad (26)$$

Similarly, the number of moles leaving the membrane, $M_d(t)$, is directly related to the pressure rise:

$$M_d(t) = \frac{V_d}{RT} \int_0^t \Delta p_d(t) dt = A \int_0^t J(L, t) dt \quad (27)$$

Fig. 4 presents the plots $M_0(t)$ and $M_d(t)$ using the experimentally measured Δp_0 and Δp_d shown in Fig. 2. Fig. 4 resembles Fig. 2 because M_0 and M_d are directly related to Δp_0 and Δp_d . However, unlike Fig. 2, the slopes $M_0(t)$ and $M_d(t)$ at a steady state are the same so that the respective linear portions of the curves are parallel. It is because in Eqs. (26) and (27) rely on the pressure decay and the pressure rise corrected by the respective upstream (V_0) and downstream (V_d) volumes. The gas accumulation in the membrane as a function of time, $M(t)$, is therefore:

$$M(t) = M_0(t) - M_d(t) \quad (28)$$

The M_∞ required to calculate S using Eq. (23) is represented by the vertical distance between $M_0(t)$ and $M_d(t)$ at a steady state. It is also important to note that the horizontal distance between $M_0(t)$ and $M_d(t)$ at a steady state represents the difference between the downstream and the upstream time lags. Because $M_0(t)$ and $M_d(t)$ are determined from the respective pressure decay (Eq. (26)) and pressure rise (Eq. (27)) responses, the resulting θ_0 and θ_d in Fig. 4 are identical, as they should, as those in Fig. 2, and their difference depends on D :

$$\theta_d - \theta_0 = \frac{L^2}{2D} \quad (29)$$

The diffusivity of N_2 from Eq. (29) represents an arithmetic average diffusivity determined from Eqs. (1) and (16).

Fig. 5 presents $M(t)$ as a function of time determined from Fig. 4. As the time increases, $M(t) \rightarrow M_\infty$, which is identical to the vertical distance between $M_0(t)$ and $M_d(t)$ at a steady state. The value of M_∞ in Fig. 5 is obtained from the arithmetic average of $M(t)$ after the system reaches the steady state at approximately 200 s. It is important to note that 200 s corresponds to four times the downstream time lag, consistent with a

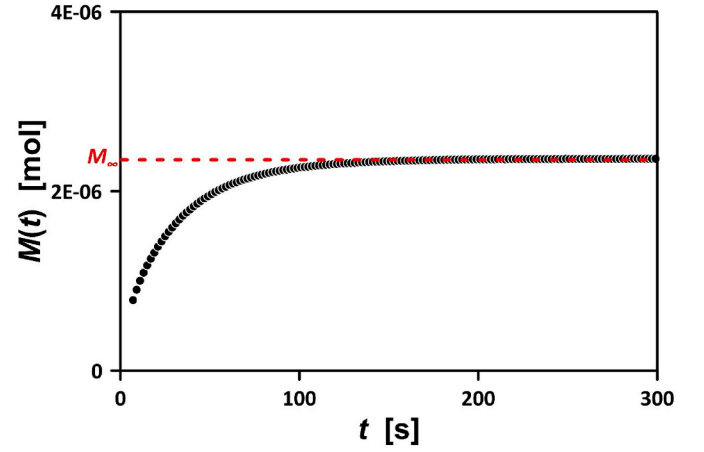


Fig. 5. Gas accumulation in the membrane as a function of time - determined from $M_d(t)$ and $M_0(t)$ presented in Fig. 4. The N_2 permeation experiment in the CV system using a PDMS membrane of thickness 0.610 ± 0.051 mm at 850 Torr and 30°C .

rule of thumb for reaching the steady state permeation [24]. According to Fig. 5 $M_\infty = 2.35 \times 10^{-6}$ mol, the corresponding solubility coefficient from Eq. (23) is $S = 5.38 \text{ mol/m}^3 \cdot \text{Pa}$.

Having the experimental value of M_∞ allows the transformation of Eq. (21) into Eq. (25), as shown in Fig. 6. It is evident that there is a linear portion in Fig. 6. The slight upward deviation at the beginning is due to Eq. (24) relying on a single-term approximation, which is not valid at short times. Additionally, there is a downward deviation from linearity when $Fo > 0.35$. As the steady state is approached, the left-hand side of Eq. (24) approaches zero. However, the limited resolution of DPT1 results in a systematic error in $M_\infty - M(t)$, which becomes evident when the difference approaches zero. Nevertheless, there is a sufficiently long period in Fig. 6 during which $M_{in}(t)$ is a linear function of t with a slope of $\sigma = -0.031 \text{ s}^{-1}$, corresponding to a diffusivity $D = 1.14 \times 10^{-9} \text{ m}^2/\text{s}$.

5. Discussion

Monitoring the pressure decay in addition to the conventional pressure rise during the time lag experiments offers three additional

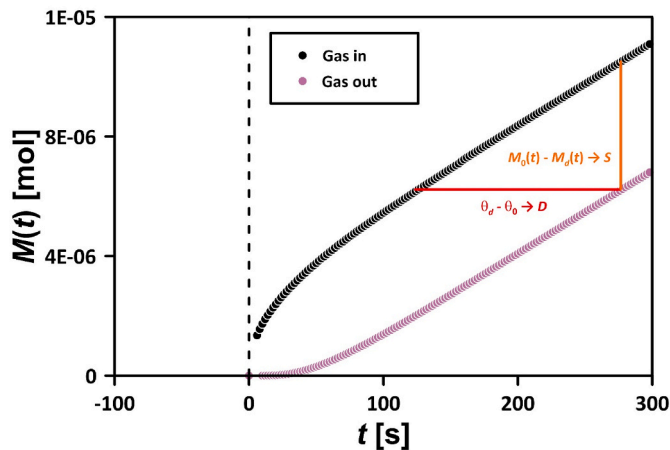


Fig. 4. The amount of gas entering and leaving the membrane as a function of time determined from $M_0(t)$ and $M_d(t)$ shown in Fig. 4 for the pressure decay and rise in Fig. 2. The N_2 permeation experiment in the CV system using a PDMS membrane of thickness 0.610 ± 0.051 mm at 850 Torr and 30°C .

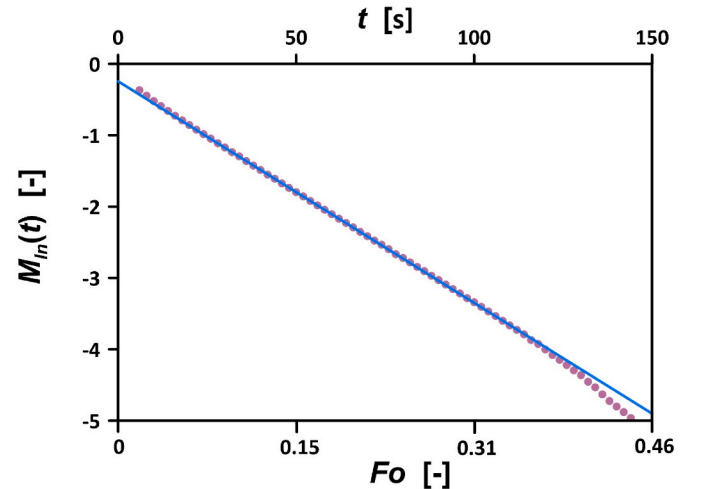


Fig. 6. Transformation of the results presented in Fig. 5 using Eq. (25). The Fourier number in the lower x-axis is calculated using $D = 1.14 \times 10^{-9} \text{ m}^2/\text{s}$ determined using the slope of the linear portion of the data in this figure. The N_2 permeation experiment in the CV system used a PDMS membrane with a thickness of 0.610 ± 0.051 mm at 850 Torr and 30°C .

methods for characterizing gas separation membranes. The first and most obvious, which we will refer to as Method 2 (where Method 1 is the time-lag method), is the direct application of the pressure decay, which allows determining D from the upstream time lag and P from the linear portion of the pressure decay, mimicking the conventional membrane characterization using the pressure rise, i.e., Method 1. The second, the two-slope method, which we will refer to as Method 3, uses the linear portion of pressure decay and the early slope of the pressure decay in the square root-of-time domain. The ratio of the two slopes depends only on the diffusion coefficient. The permeability coefficient in this method is the same as in the first method. These two new methods solely depend on pressure decay. If P , D , and S can be determined solely based on pressure decay using Methods 2 and 3, the permeate side of the membrane could be continuously evacuated, better adhering to one of the two boundary conditions on which the time-lag method relies. Alternatively, the permeate side of the membrane could be connected to an online mass spectrometer, which would be advantageous if the feed were a gas mixture rather than a single gas. Finally, the combined method (Method 4), which relies on both pressure rise and decay, allows for the determination of gas accumulation in the membrane as a function of time. This, in turn, allows the direct determination of the diffusivity and solubility coefficients in a way similar to the dynamic adsorption/desorption experiments used to characterize adsorbents. The resulting P is the product of directly determined D and S .

Table 1 summarizes the P , D and S values using the abovementioned methods. Methods 1–3 rely on the fundamental correlation of the solution diffusion mechanism, i.e.

$$P = DS \quad (30)$$

In other words, S in Methods 1–3 represents the ratio of the respective P and D values. On the other hand, S in Method 4 is determined independently and does not rely on Eq. (30). Because Method 4 utilizes the pressure rise and pressure decay results, the reported P in Table 1 is the arithmetic average from Methods 1 and 2.

In addition to the values listed in Table 1, one could also report the average D and S from Methods 1 and 2: $D = 1.26 \times 10^{-9} \text{ m}^2/\text{s}$ and $S = 5.62 \times 10^{-5} \text{ [mol/m}^3 \text{ Pa]}$. Also, using the product of D and S determined from Method 4, one could report $P = 186 \text{ [Barrer]}$. The latter is 13 % smaller than the average P (214 Barrer). This discrepancy is primarily because the D from Method 4 is 10 % smaller than the average D from Methods 1 and 2. As indicated by Eq. (25), due to the logarithmic nature of the right-hand of this equation, a small error in M_∞ , such as 1 %, could translate into a larger error in the slope in Fig. 6, from which D is evaluated in Method 4.

On the other hand, a 1 % error in M_∞ would result in a 1 % error in S from Method 4. More importantly, the S from Method 4 does not rely on Eq. (30). As previously mentioned, the evaluation of S using Method 4 resembles the determination of S using a barometric method [11]. However, there are some fundamental differences. First, M_∞ in Method 4 represents a steady-state rather than an equilibrium value. At a steady state, for a given feed pressure, while the permeate side is a vacuum, the membrane is “filled” to 50 % of its capacity because of the linear

concentration profile. If an impermeable layer covered the permeate side of the membrane, the membrane could be filled to 100 % of its capacity, which would resemble a typical sorption experiment. However, it can be shown that filling the remaining 50 % of the capacity would take a much longer time than reaching the steady state. It is important to acknowledge that in a sorption experiment, a sample (i.e., the membrane) would be exposed to the same pressure conditions on both the feed and permeate sides. As a result, the characteristic length for diffusion inside the membrane would be half of its thickness rather than the entire thickness in the permeation experiment. Still, because of a decreasing concentration gradient for diffusion as the gas accumulates in the sample, reaching the equilibrium would take much longer than reaching a steady state in the permeation cell [25].

Another advantage of determining the solubility coefficient using Method 4 in our CV system is the resolution of the measurement. PDMS or, in general, polymeric membranes do not have high-capacity absorption capacities. Therefore, evaluating the gas solubility coefficient in conventional gas sorption experiments using gravimetric/barometric/volumetric techniques would require a larger sample, which could increase the characteristic length for diffusion and, consequently, the time to reach the equilibrium concentration in the tested sample [11]. Also, as shown by Eq. (23), knowing the experimental M_∞ , requires accurate values of p_0 , L and A to determine S . There is practically no uncertainty in these values when a membrane is tested in a CV system. The accurate knowledge of L , or, more generally, the characteristic length for diffusion, is even more critical in determining D using Method 4 (Eq. (25)), as the slope in Fig. 6 is inversely proportional to the square of membrane thickness.

In this work, we have demonstrated the “multiplication” of membrane characterization methods when the pressure decay and the pressure rise are monitored simultaneously in a conventional time-lag experiment. The importance of using multiple methods (e.g., integral and differential) for analyzing the permeation data from a given time lag experiment was postulated by Beckman et al. [26]. This approach helps detect anomalous diffusion effects. For the rubbery polymers such as PDMS, P , D , and S should be independent of pressure. However, these transport coefficients become pressure-dependent in glassy polymers. Moreover, the more complex membrane materials, including glassy polymer membranes, require additional kinetic and thermodynamic parameters to characterize their performance fully. In this case, the “multiplication” of membrane characterization methods using simultaneous monitoring of pressure rise and decay could be advantageous and necessary.

6. Conclusions

The time-lag method, the most common way to evaluate membranes D , P and S , is widely used to design and optimize advanced membrane materials in gas separation. The conventional time-lag method, which focuses on the downstream pressure profile, evaluates solubility from the ratio of permeability and diffusivity. In this work, we have provided the theoretical basis and experimentally demonstrated the direct determination of D , P , and S from a single time-lag gas permeation test by simultaneously measuring the upstream pressure decay and downstream pressure rise in a new generation of a CV system.

The simultaneous monitoring of pressure decay and pressure rise allows the evaluation of gas accumulation in the membrane as a function of time, and the steady-state gas accumulation allows the direct determination of S . Moreover, the time-dependent gas accumulation in the membrane allows direct determination of D . Monitoring pressure decay during the time lag experiment also enables a new way of determining D from the ratio of two slopes, the early slope of pressure decay in the square root of time domain and the steady state pressure decay. In addition, the entire pressure decay curve mimics the conventional pressure rise curve from which P and D can be directly evaluated. The P , D , and S of N_2 in the PDMS membrane using these characterization

Table 1

The summary of N_2 transport coefficients in a PDMS membrane of thickness $0.610 \pm 0.051 \text{ mm}$ determined from the single gas permeation (time-lag) experiment at 850 Torr and 30 °C.

Method	P [Barrer]	D [m^2/s]	S [$\text{mol}/\text{m}^3 \text{ Pa}$]
1. Conventional pressure rise	207	1.26×10^{-9}	$5.40 \times 10^{-5} \text{ }^a$
2. Direct pressure decay	221	1.25×10^{-9}	$5.83 \times 10^{-5} \text{ }^a$
3. Pressure decay two-slopes	221 ^b	1.28×10^{-9}	$5.70 \times 10^{-5} \text{ }^a$
4. Combined pressure rise & decay	214 ^c	1.14×10^{-9}	5.38×10^{-5}

^a Calculated from $P = DS$.

^b From the pressure decay slope – Method 2.

^c The arithmetic average from Methods 1 and 2.

methods were similar, which provided validation of the new methods proposed in this work.

Multiple methods for determining P , D , and S in rubbery polymers might be unnecessary. Still, having several independent values of transport coefficients for a given membrane provides more confidence. However, when it comes to more complex membrane materials, in which Henry’s law is not applicable, including glassy polymers, mixed matrix materials, and membranes with anomalous diffusion effects, the conventional time-lag method might be insufficient, and the new methods proposed in this work may become indispensable.

CRediT authorship contribution statement

Zheng Cao: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Jules Thibault:** Writing – review & editing, Supervision, Software, Methodology, Investigation,

Conceptualization. **Boguslaw Kruczek:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge the financial support provided by the Natural Science and Engineering Research Council (NSERC) Canada, Discovery Grant (DG), grant number: 04443.

Nomenclature

Symbols	
A	Membrane area [m ²]
c	Gas concentration in the membrane [mol/m ³]
D	Diffusivity coefficient [m ² /s]
$ Fo$	Fourier number [–]
L	Membrane thickness [m]
$M(t)$	Gas accumulation in the membrane as a function of time [mol]
$M_{ln}(t)$	Logarithmic gas accumulation defined by Eq. (25) [–]
M_{∞}	Gas accumulation in the membrane at the steady state [mol]
P	Permeability coefficient [Barrer or mol/s m Pa]
p	Pressure [Torr or Pa]
R	Universal gas constant [J/mol K]
S	Solubility coefficient [mol/m ³ Pa]
T	Absolute temperature [K]
t	Time [s]
V	Volume [m ³]

Greek symbols	
θ	Time lag [s]
Δp	Pressure decay/pressure rise [Torr or Pa]
σ	Slope defined by Eq. (15) [1/s]
Σ_t	Slope in the square-root-of-time domain defined by Eq. (17) [Torr/s ^{1/2} or Pa/s ^{1/2}]
Σ_{ss}	Slope in the time domain defined by Eq. (18) [Torr/s or Pa/s]

Subscripts	
d	Downstream
0	Upstream

Data availability

Data will be made available on request.

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